7 **(a)** Figure 7.1 shows a simplified diagram of a loudspeaker.

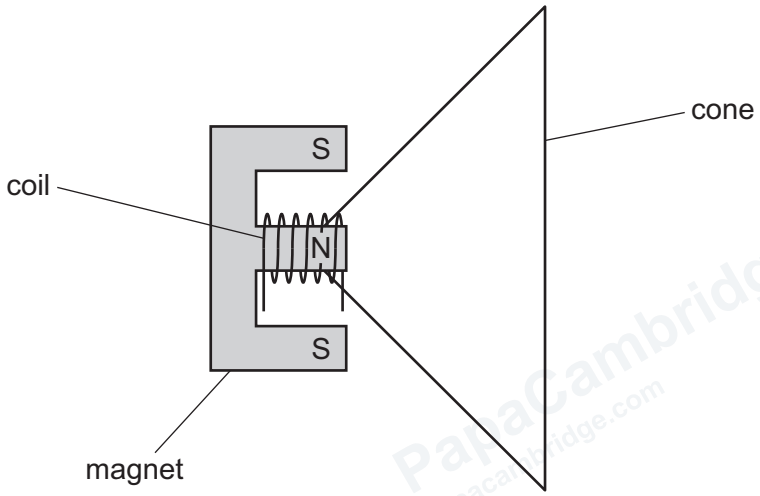

Figure 7.1

(i) Complete the sentences to describe and explain how a loudspeaker produces sound.

A changing magnetic field is produced by the ..... in the coil.

This magnetic field interacts with the magnetic field due to

the ..... .

This causes the cone to vibrate. The vibration of the .....

transfers energy to a student as sound.

[3]

(ii) A sound of frequency 440 Hz is produced by the loudspeaker. The speed of the sound is 340 m / s.

Calculate the wavelength of this sound. State the unit.

wavelength = ... unit ..... [2]

(b) Define 'ultrasound'.

.....

..... [1]

[Total: 6]

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